

Foundations of Hidden Markov model

[Hidden Markov model]

$$P(O | \lambda) = \sum_Q P(O | Q, \lambda) P(Q | \lambda)$$

1.0 Content Block

Most likely explanation The task, unlike the previous two, asks about the joint probability of the entire sequence of hidden states that generated a particular sequence of observations (see illustration on the right). This task is generally applicable when HMM's are applied to different sorts of problems from those for which the tasks of filtering and smoothing are applicable. An example is part-of-speech tagging, where the hidden states represent the underlying parts of speech corresponding to an observed sequence of words. In this case, what is of interest is the entire sequence of parts of speech, rather than simply the part of speech for a single word, as filtering or smoothing would compute. This task requires finding a maximum over all possible state sequences, and can be solved efficiently by the Viterbi algorithm.

2.0 Content Block

History Hidden Markov models were described in a series of statistical papers by Leonard E. Baum and other authors in the second half of the 1960s. One of the first applications of HMMs was speech recognition, starting in the mid-1970s. From the linguistics point of view, hidden Markov models are equivalent to stochastic regular grammar. In the second half of the 1980s, HMMs began to be applied to the analysis of biological sequences, in particular DNA. Since then, they have become ubiquitous in the field of bioinformatics.

3.0 Content Block

Statistical significance For some of the above problems, it may also be interesting to ask about statistical significance. What is the probability that a sequence drawn from some null distribution will have an HMM probability (in the case of the forward algorithm) or a maximum state sequence probability (in the case of the Viterbi algorithm) at least as large as that of a particular output sequence? When an HMM is used to evaluate the relevance of a hypothesis for a particular output sequence, the statistical significance indicates the false positive rate associated with failing to reject the hypothesis for the output sequence.

4.0 Content Block

Learning The parameter learning task in HMMs is to find, given an output sequence or a set of such sequences, the best set of state transition and emission probabilities. The task is usually to derive the maximum likelihood estimate of the parameters of the HMM given the set of output sequences. No tractable algorithm is known for solving this problem exactly, but a local maximum likelihood can be derived efficiently using the Baum–Welch algorithm or the Baldi–Chauvin algorithm. The Baum–Welch algorithm is a special case of the expectation-maximization algorithm. If the HMMs are used for time series prediction, more sophisticated Bayesian inference methods, like Markov chain Monte Carlo (MCMC) sampling are proven to be favorable over finding a single maximum likelihood model both in terms of accuracy and stability. Since MCMC imposes significant computational burden, in cases where computational scalability is also of interest, one may alternatively resort to variational approximations to Bayesian inference, e.g. Indeed, approximate variational inference offers computational efficiency comparable to expectation-maximization, while yielding an accuracy profile only slightly inferior to exact MCMC-type Bayesian inference.

5.0 Content Block

Smoothing This is similar to filtering but asks about the distribution of a latent variable somewhere in the middle of a sequence, i.e. to compute $P(x(k) \mid y(1), \dots, y(t))$ for some $k < t$. From the perspective described above, this can be thought of as the probability distribution over hidden states for a point in time k in the past, relative to time t . The forward-backward algorithm is a good method for computing the smoothed values for all hidden state variables.

6.0 Content Block

A hidden Markov model (HMM) is a Markov model in which the observations are dependent on a latent (or hidden) Markov process (referred to as X). An HMM requires that there be an observable process Y whose outcomes depend on the outcomes of X in a known way. Since X cannot be observed directly, the goal is to learn about state of X by observing Y . By definition of being a Markov model, an HMM has an additional requirement that the outcome of Y at time $t = t_0$ must be "influenced" exclusively by the outcome of X at $t = t_0$ and that the outcomes of X and Y at $t < t_0$ must be conditionally independent of Y at $t = t_0$ given X at time $t = t_0$. Estimation of the parameters in an HMM can be performed using maximum likelihood estimation. For linear chain HMMs, the Baum–Welch algorithm can be used to estimate parameters. Hidden Markov models are known for their applications to thermodynamics, statistical mechanics, physics, chemistry, economics, finance, signal processing, information theory, pattern recognition—such as speech recognition, handwriting recognition, gesture recognition, part-of-speech tagging, musical score following, partial discharges and bioinformatics.

7.0 Content Block

$$P(Y_{t_0} \in A \mid \{X_t \in B_t\}_{t \leq t_0}) = P(Y_{t_0} \in A \mid X_{t_0} \in B_{t_0})$$

$$\{\mathbf{P}\}(Y_{t_0} \in A \mid \{X_t \in B_t\}_{t \leq t_0}) = \{\mathbf{P}\}(Y_{t_0} \in A \mid X_{t_0} \in B_{t_0}),$$
for every t_0 , every Borel set A , and every family of Borel sets $\{B_t\}_{t \leq t_0}$.

8.0 Content Block

In this piece of code, `start_probability` represents Alice's belief about which state the HMM is in when Bob first calls her (all she knows is that it tends to be rainy on average). The particular probability distribution used here is not the equilibrium one, which is (given the transition probabilities) approximately {'Rainy': 0.57, 'Sunny': 0.43}. The `transition_probability` represents the change of the weather in the underlying Markov chain. In this example, there is only a 30% chance that tomorrow will be sunny if today is rainy. The `emission_probability` represents how likely Bob is to perform a certain activity on each day. If it is rainy, there is a 50% chance that he is cleaning his apartment; if it is sunny, there is a 60% chance that he is outside for a walk.